

Sri Lanka Institute of Information Technology



IT3091 – Machine Learning
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Intelligent Employee Task Allocation Recommendation System Using Machine Learning

Client: Cygnus One (Pvt) Ltd
Track: Industry Explorer Track

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1.Introduction

1.1 Project Overview

This project is conducted under the Industry Explorer Track of the Machine Learning module in collaboration with Cygnus One (Pvt) Ltd, a technology solutions company operating in Sri Lanka.

Organizations that manage multiple employees, projects, and tasks need to allocate tasks to employees who possess the appropriate experience, skills, and knowledge. When the number of employees and tasks increases, manually identifying the most suitable employee for each task can become time consuming and dependent on individual managerial judgement.

Employee position, department, project, supervisor, task details, task descriptions, job length, and prior employee experience will all be analyzed by the proposed system. After then, machine learning methods will be applied to find trends in past task assignments and produce suggestions for future tasks.

The system will identify and prioritize the workers who are most suited for a new assignment when a manager offers information about it. The system is designed to serve as a decision-support tool, with the manager making the ultimate allocation choice.

1.2 Client Overview

Cygnus One (Pvt) Ltd is a technology solutions company that focuses on ERP implementation, application development, business consultation, AI, automation, and digital transformation. The company has more than ten years of industry experience and identifies Odoo ERP as one of its core areas of expertise.

Cygnus One provides and customizes Odoo solutions covering areas such as Sales & CRM, Finance & Accounting, Inventory & Warehouse, Human Resources, Project Management, and Website & eCommerce. Its Project Management offering includes task assignments, deadlines, milestones, and employee allocation.

The company also describes its R&D activities as involving AI automation and Odoo custom integrations. Departments include Business Solutions, Client Support, Research & Development, Human Resources, Sales & Marketing, and Administration to make up its organizational structure.

These fields offer a pertinent industrial context for creating a machine-learning solution pertaining to job distribution, operational decision support, and staff skills.

1.3 Project Motivation

The motivation for this project is the challenge of assigning the most suitable employee to a new task using available historical information.

Currently, task allocation may require managers to manually consider several factors, including:

- employee position
- department
- previous project experience
- previous task assignments
- task description
- task duration
- supervisor
- employee experience
- similarity between the new task and tasks previously completed.

As the number of employees, projects, and tasks increases, considering all of these factors manually can become difficult. Therefore, this project aims to use historical organizational data to provide a data driven employee recommendation mechanism. The motivation is not to replace managers, but to provide them with additional evidence when making allocation decisions.

2. Industry Explorer Track

2.1 Reason for selecting Industry Explorer

The project has been selected under the Industry Explorer Track because it addresses a real world organizational problem in collaboration with an actual industry client, Cygnus One (Pvt) Ltd.

The Industry Explorer approach allows the team to:

1. Understand the client's existing task-allocation process.
2. Identify the information currently used when allocating tasks.
3. Obtain and analyze relevant historical data.
4. Develop a machine learning solution based on the actual business context.
5. Translate the model results into a practical recommendation for the organization.

The project therefore connects machine learning techniques with a real operational decision rather than treating model development as an isolated technical exercise.

2.2 Expected Industry Engagement

The project team will maintain regular communication with Cygnus One to ensure that the proposed system addresses a real business need. The engagement will include discussing the current task allocation process, employee and project information, available historical data, and task assignment factors. The client will also provide feedback during model validation to assess whether the recommendations are practically meaningful. Finally, the completed

recommendation system, its limitations, and potential usefulness in the real working environment will be discussed with the client.

2.3 Expected Business Value

- **Faster Decisions & Operational Efficiency:** Replaces manual candidate screening with instant, data-driven recommendation lists, streamlining the assignment process.
- **Smarter Knowledge & Experience Match:** Systematically leverages historical project data and prior task experience to match the right employee to the right role.
- **Consistency & Objectivity:** Eliminates subjective bias by standardizing task assignments against established criteria and model data.

3. Business Lens

3.1 Proposed Primary Lens

The primary business lens for this project is Operational Efficiency and Workforce Allocation. This lens focuses on improving how available employees are allocated to organizational tasks.

The central decision supported by the project is Which employee is most suitable for a given task?

3.2 Rationale

This lens was selected because the system directly addresses an operational resource-allocation challenge by using historical data to automate candidate ranking, thereby significantly reducing manual evaluation effort. By standardizing the allocation process and systematically leveraging past employee experience, the project improves decision-making speed, enhances task allocation efficiency, and ensures consistent, optimal workforce utilization.

3.3 Possible Secondary Lens

A possible secondary lens is Employee Capability and Skill Utilization . This lens considers how employee experience and previous task history can be used to identify employees with relevant capabilities for new tasks.

However, this is a secondary lens because the main objective is not to evaluate employees' overall performance. The purpose is Use employee capability information to improve task allocation.

3.4 Lens Selection Criteria

- **Business & Decision Value:** Solves a genuine organizational problem and directly supports practical managerial decisions.
- **Data & Technical Viability:** Relies on accessible, client-approved historical data with patterns suitable for machine learning.

- **Measurability & Governance:** Delivers outputs that can be evaluated with defined metrics while maintaining strict ethical standards and fair employee treatment.

4. Proposed ML/Data Mining Task

4.1 Task Selection Approach

The proposed machine-learning task is Employee Task Suitability Prediction and Ranking. The system will use historical employee–task assignment data to identify relationships between employee experience, project information, and task characteristics. For a new task, it will estimate employee suitability and generate a ranked list of suitable employees. The final task formulation will be confirmed based on the available dataset, client defined suitability criteria, data quality, historical assignments, and the client's expected recommendation output.

4.2 ML Approach

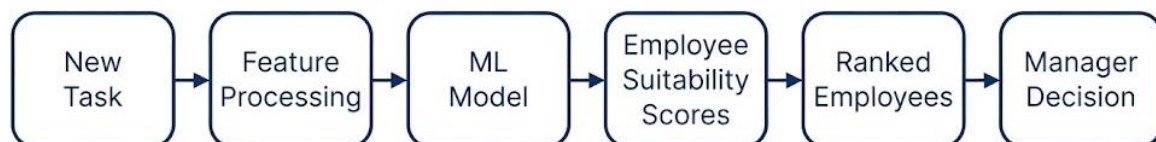
The project will primarily focus on **employee suitability scoring and ranking**.

- **Scoring & Classification:** Combines predictive models to calculate employee suitability scores/probabilities and determine binary fit where applicable.
- **Text Similarity (NLP):** Utilizes NLP to match new task descriptions against employees' past project documentation and task history.
- **Ranking & Evaluation:** Ranks candidates by overall suitability score, selecting the optimal model based on predictive performance and practical business utility.

4.3 Expected Output

The main expected output is a ranked list of the most suitable employees for a new task.

The proposed system workflow is:



The system will support managers in making task-allocation decisions, while the final task assignment will remain under human supervision.

5. Proposed Workflow

The proposed system will develop a machine learning based employee task allocation recommendation system. The system will learn from historical employee and task allocation data and, when a new task is provided, recommend and rank the most suitable employees.

5.1 Business Problem Identification

The project will begin by understanding Cygnus One's current employee task allocation process, including the factors considered, existing challenges, and the characteristics of a suitable employee. Based on this understanding, the system will aim to recommend and rank suitable employees for a new task using historical employee task data and relevant employee and task characteristics. The system will serve as a decision support tool for managers, rather than replacing human decision-making.

5.2 Data Collection

The team will collaborate with the client to acquire approved historical data, including employee profiles, project and task descriptions, timelines, and past assignment records. Pending client approval and data availability, this core dataset may be further enriched with supplemental variables such as specific employee skills, technology stacks, task priorities, completion statuses, and current workloads. The dataset will be anonymized where necessary to protect employee privacy.

5.3 Data Understanding & EDA

- **Data Quality Assessment:** Inspects dataset size, structure, missing values, and duplicates to ensure foundational data health.
- **Pattern & Distribution Analysis:** Examines historical allocation trends, employee-task frequencies, and distributions across departments, projects, and task durations.
- **Modeling Preparation:** Proactively uncovers key predictive relationships and flags potential data issues to guide effective model development.

5.4 Preprocessing

- The dataset will be cleaned and formatted through standard steps, including handling missing values, removing duplicates, encoding/standardizing variables, processing text, and splitting the data into training and testing sets.

Data Leakage Prevention

Data leakage will be specifically investigated.

Features that would only be known after a task has been assigned or completed will not be used to predict the employee for that task.

5.5 Feature Engineering

- **Data Transformation:** Converts raw employee, project, and task data into structured machine learning features.
- **Text Processing:** Utilizes TF-IDF for lightweight, interpretable task description analysis, with the potential to explore advanced text embeddings if justified by dataset size and requirements.
- **Feature Selection:** The final set of features will be determined by Exploratory Data Analysis (EDA) findings, data availability, and specific client needs

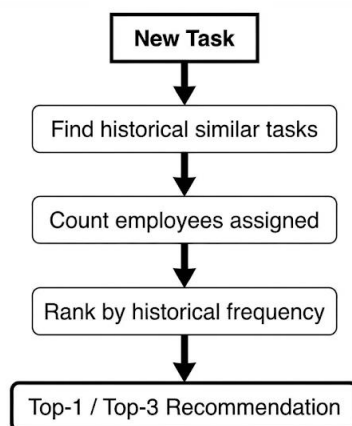
5.6 Baseline Establishment

Before training machine-learning models, a non ML baseline will be established.

The baseline will provide a reference point against which the machine-learning models can be compared. It is not intended to guarantee a minimum final model accuracy rather it establishes the performance that a simple allocation strategy can achieve.

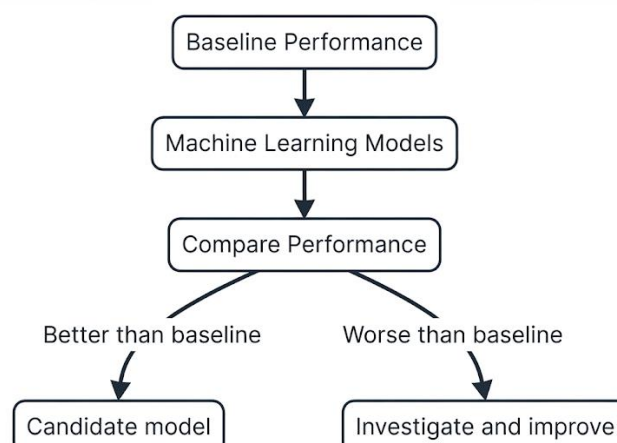
A suitable baseline for this project could be a historical frequency based recommendation.

For example, the system could recommend employees who have historically received the highest number of similar tasks.



The baseline performance will be measured using the same appropriate test data and evaluation metrics used for the ML models.

This provides a meaningful benchmark:



A trained model will only be considered useful if it provides sufficient improvement over the baseline or provides additional business value such as better ranking, interpretability, or workload balancing.

5.7 Model Development

Multiple machine learning models will be trained and compared rather than relying on a single algorithm.

The proposed models are:

5.7.1 Logistic Regression

Logistic Regression will be used as a simple and interpretable machine learning model. It provides a useful comparison against more complex models and can produce probability estimates that can subsequently be used to rank employees.

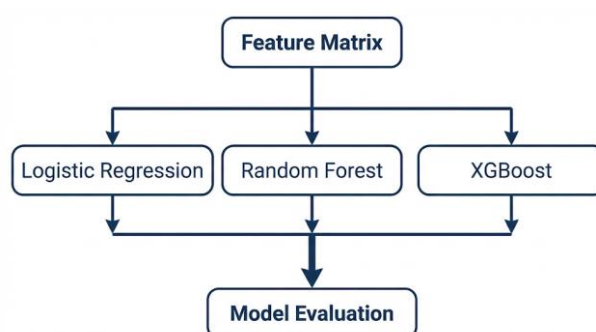
5.7.2 Random Forest

Random Forest will be used as a nonlinear tree based model capable of handling interactions between employee, task, project, and numerical features.

It can also provide feature importance information that may help explain the recommendation process.

5.7.3 XG Boost

XG Boost will be used as the primary advanced tree based candidate. It is well suited to structured/tabular data and can model complex relationships between employee characteristics, task characteristics, and historical experience.



The final model will not be selected solely based on one metric. Performance, interpretability, computational requirements, generalization, ranking quality, and business usefulness will also be considered.

5.8 Evaluation

- **Testing Strategy:** Models will be evaluated on a held out test set, starting with default configurations to compare against a baseline.
- **Classification Metrics:** Standard performance metrics (Accuracy, Precision, Recall, F1-score, ROC/PR-AUC) will be considered based on the final task formulation.
- **Ranking-Oriented Focus:** Primary emphasis will be placed on ranking evaluation, specifically Top K Hit Rate (Top 1, Top 3, Top 5), to ensure the system directly supports the business requirement of providing ranked employee recommendations.

5.9 Hyperparameter Optimization Using Optuna

- **Framework & Models:** Optuna will optimize the top candidate models, specifically tuning key parameters for tree-based architectures like XG Boost (depth, learning rate, subsampling) and Random Forest (tree count, split/leaf limits).
- **Validation & Objective:** Optimization will prioritize ranking-oriented validation metrics (such as Top-K Hit Rate) combined with cross-validation to ensure reliable performance across data splits.
- **Evaluation Integrity:** The test dataset will remain completely isolated and untouched throughout the optimization process, reserved strictly for final evaluation.

5.10 Business Recommendation

The final stage will translate the model results into a practical business recommendation.

- **Model Justification:** Details the rationale for the selected model, its performance improvement over the baseline, and the key factors driving its recommendations.
- **Operational Value:** Outlines the practical benefits of ranked recommendations and how they specifically support managerial decision-making.
- **Risk Management & Scope:** Highlights model limitations and the risks of historical bias, firmly establishing the system as an assistive decision support tool that requires human oversight rather than an autonomous allocator.

6.Team Responsibilities

Member	Role	Main Contribution	Specific Tasks / Activities
IT24100677 – Samarathunga S.D.D	Data Engineering & EDA Lead	Responsible for preparing and understanding the raw dataset and identifying meaningful patterns before model development.	• Acquire and organize the dataset • Understand dataset structure and variables • Perform data cleaning • Conduct Exploratory Data Analysis (EDA) • Perform statistical analysis • Create EDA visualizations • Identify important patterns, trends, and relationships in the data
IT24100108 – Sanjana B.A	Data Preprocessing & Feature Engineering Lead	Responsible for transforming the cleaned data into a suitable format and creating meaningful features for the ML models.	• Handle missing values • Perform categorical encoding • Apply numerical preprocessing • Preprocess task-description text • Perform feature engineering • Develop employee-task relationship features • Check for data leakage • Prepare training and testing datasets
IT24100307 – Bandara I.G.C	Machine Learning Model Development Lead	Responsible for developing, training, tuning, and comparing machine-learning models for employee-task suitability prediction.	• Develop baseline model • Implement candidate ML models • Train machine-learning models • Perform hyperparameter tuning • Apply cross-validation • Track model experiments and results • Compare model performance • Select the most promising model for further evaluation
IT24101887 – Faizam M.F.M	Model Evaluation, Explainability & Recommendation Lead	Responsible for evaluating the developed models and converting model predictions into understandable employee recommendations.	• Evaluate model performance • Calculate appropriate evaluation metrics • Conduct error analysis • Compare final model results • Analyze feature importance • Generate employee suitability rankings • Generate Top-N employee recommendations • Interpret results from a business perspective

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